Surface Phenomena & Catalysis
Week 2: Reactor Design & Catalyst Characterization

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Course Roadmap

Week	Chapter	Topic
1	Ch 1–2	Bridge Module + Computational Perspectives
2	**Ch 3**	**Reactor Design & Catalyst Characterization**
3	Ch 4	Transport Limitations
4	Ch 5	Selectivity Engineering
5	Ch 6	Zeolites & Carbon Nanotubes

From rate law to reactor sizing, and the techniques that reveal catalyst structure.

Today's Agenda

Part I: Reactor Design *~30 min*

Batch kinetics, half-life diagnostics, TOF/TON, Sabatier principle, CSTR vs PFR design equations

Part II: Catalyst Characterization *~40 min*

BET, TPD, TPR/TPO, XRD, XPS, TEM, CO-FTIR — worked examples & integration

Part III: Summary & Self-Study *~10 min*

Key equations + NB3 self-study guide + looking ahead to Week 3

Lecture notes: Chapter 3 — Notebook: NB3

Outline

1 Part I: Reactor Design

2 Part II: Catalyst Characterization

3 Summary & Self-Study

Batch Reactor: The Master Equation

Mole balance on species A:

$$-\frac{\mathrm{d}C_A}{\mathrm{d}t} = r(C_A)$$

- Closed system, constant volume
- $r > 0$ by convention (consumption rate)
- The **form** of $r(C_A)$ determines the method

Strategy

1. Assume a rate law: $r = k\,C_A^n$
2. Integrate the mass balance
3. Plot data in the linearised form
4. Best straight line $\Rightarrow$ correct n

Integrated Rate Laws — Three Cases

Order n	Integrated Form	Linear Plot
0	$C_A = C_{A0} - kt$	C_A vs t
1	$\ln C_A = \ln C_{A0} - kt$	$\ln C_A$ vs t
2	$1/C_A = 1/C_{A0} + kt$	$1/C_A$ vs t

Key diagnostic

Plot all three transforms; the one that gives a **straight line** reveals the reaction order. Always confirm with a **residual plot** — a "good" R^2 can mask systematic curvature.

Linearisation Strategy for Kinetic Analysis

Two complementary methods:

Integral method

1. Assume an order n
2. Plot the corresponding transform
3. Check linearity (R^2 or residuals)

Best for: high-quality $C(t)$ data.

Differential method (initial rates)

1. Measure r_0 at several C_{A0}
2. Plot $\ln r_0$ vs $\ln C_{A0}$
3. Slope $= n$, intercept $= \ln k$

Best for: complex mechanisms or multiple reactants.

Always inspect residuals—a "good" R^2 can hide systematic error.

Half-Life as a Reaction Order Diagnostic

General formula for $r = k\,C_A^n$:

$$t_{1/2} = \frac{2^{n-1}-1}{(n-1)\,k\,C_{A0}^{n-1}} \qquad (n \neq 1)$$

n	$t_{1/2}$ expression	Depends on C_{A0}?
0	$C_{A0}/(2k)$	$\propto C_{A0}$
1	$\ln 2/k$	**No!**
2	$1/(k\,C_{A0})$	$\propto 1/C_{A0}$

Quick diagnostic

- Dilute the reactant by half
- Measure $t_{1/2}$ again
- If $t_{1/2}$ **unchanged** $\Rightarrow$ 1st order
- If $t_{1/2}$ **doubles** $\Rightarrow$ 2nd order
- If $t_{1/2}$ **halves** $\Rightarrow$ 0th order

Worked Example: Determining Reaction Order

Problem

Decomposition of N_2O_5 in CCl_4 at 45 °C gives the data below. Determine the reaction order.

t (min)	0	10	20	30	40	50
C_A (mol L^{-1})	2.33	2.08	1.91	1.67	1.49	1.35
$\ln C_A$	0.846	0.732	0.647	0.513	0.399	0.300
$1/C_A$ (L mol^{-1})	0.429	0.481	0.524	0.599	0.671	0.741

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Test

$\ln C_A$ vs t is **linear** ($R^2 = 0.998$)

$\Rightarrow$ **First-order reaction**

Result

Slope $= -k = -0.0110$ min^{-1}

$t_{1/2} = \ln 2/0.0110 = 63.0$ min

(independent of C_{A0} — confirms 1st order)

Method of Initial Rates

When the integral method is insufficient: complex mechanisms, multiple reactants, or when full $C(t)$ data is hard to obtain.

Procedure:

1. Run several experiments with different C_{A0}
2. Measure the **initial rate** r_0 for each
3. For $r_0 = k\,C_{A0}^n$:

$$\ln r_0 = \ln k + n \ln C_{A0}$$

Slope $= n$, intercept $= \ln k$

When to use

- Multiple reactants (isolate each)
- Reversible reactions
- Complex product inhibition
- Catalyst deactivation

Caution

Requires **accurate** measurement of initial slopes — tangent or polynomial fit.

Turnover Frequency & Turnover Number

Rate constants describe the **reaction**; TOF describes the **catalyst**.

$$\text{TOF} = \frac{\text{moles product}}{\text{moles active sites} \times \text{time}}$$

$$\text{TON} = \text{TOF} \times t$$

- TOF: **activity** (how fast per site), units s^{-1}
- TON: **productivity** (total turnovers), dimensionless
- TON > 1: true catalysis (catalyst regenerated)

Site counting matters!

Different probe molecules give different site counts:

- CO chemisorption (Pt, Pd)
- H_2 dissociative chemisorption
- NH_3-TPD (acid sites)

Always report the method used.

Worked Example: TOF from CO Chemisorption

Problem

A 5% Pd/Al_2O_3 catalyst (0.50 g) converts cyclohexene at a rate of 2.4×10^{-4} mol s^{-1}. CO chemisorption uptake: 55 μmol g^{-1} (stoichiometry CO :Pd_{surf} = 1 :1). Calculate the TOF.

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Step 1: Surface Pd sites

$n_{\text{sites}} = 55\ \mu\text{mol/g} \times 0.50\ \text{g}$
$= 27.5\ \mu\text{mol} = 2.75 \times 10^{-5}$ mol

Step 2: TOF

$$\text{TOF} = \frac{2.4 \times 10^{-4}}{2.75 \times 10^{-5}} = \mathbf{8.7\ s^{-1}}$$

Dispersion

Total Pd: $\frac{0.05 \times 0.50}{106.42} = 2.35 \times 10^{-4}$ mol

$$D = \frac{n_{\text{surf}}}{n_{\text{total}}} = \frac{2.75 \times 10^{-5}}{2.35 \times 10^{-4}} = 0.117$$

Only 11.7% of Pd atoms are on the surface!

Sabatier Principle & Volcano Plots

Think: if you make a catalyst bind reactants **more strongly**, *does the rate always increase?*

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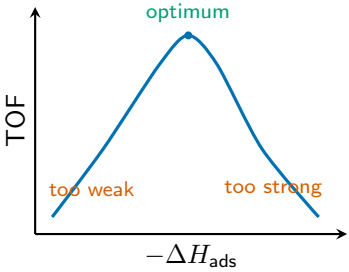
The Sabatier Principle:

> *The optimal catalyst binds reactants* **not too weakly** *(poor adsorption) and* **not too strongly** *(surface poisoning).*

- Weak binding: low $\theta \Rightarrow$ low rate
- Strong binding: high θ but slow desorption
- Optimum: intermediate ΔH_{ads}

Modern link: Nørskov DFT volcano plots screen catalysts by computed ΔE_{ads} — accelerates catalyst discovery.



Volcano plot: activity peaks at intermediate binding strength.

Flow Reactors: CSTR vs PFR

CSTR (perfectly mixed):

$$\tau = \frac{C_{A0}\,X}{r(C_{A,\text{exit}})}$$

- Algebraic: $C_{A,\text{exit}} = C_{A0}(1 - X)$
- Operates at **exit** concentration
- Lower conversion at same τ

PFR (plug flow):

$$\tau = C_{A0}\int_0^X \frac{\mathrm{d}X}{r(C_A)}$$

- ODE with $C_A = C_{A0}(1 - X)$
- Concentration decreases along reactor
- Higher conversion at same τ

Space time: $\tau = V_{\text{reactor}}/\dot{V}_0$ Contact time: $t_c = V_{\text{cat}}/\dot{V}$

CSTR vs PFR: First-Order Kinetics

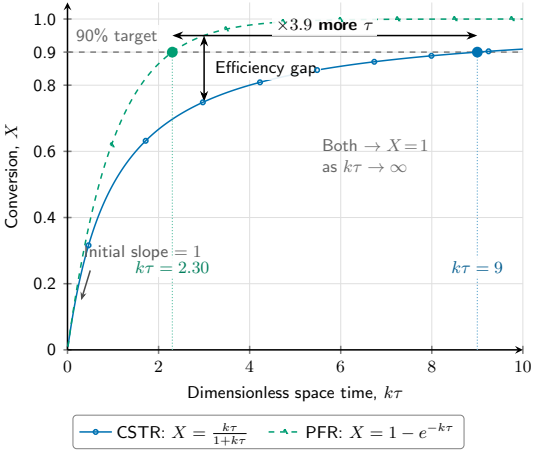
Closed-form solutions ($n = 1$):

CSTR: $X = \dfrac{k\tau}{1 + k\tau}$

PFR: $X = 1 - \exp(-k\tau)$

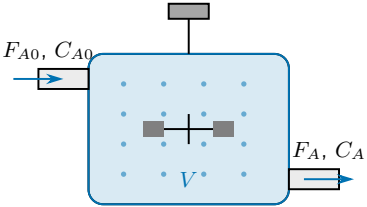
Key result

PFR **always** achieves higher X than CSTR at the same τ (for positive-order kinetics).

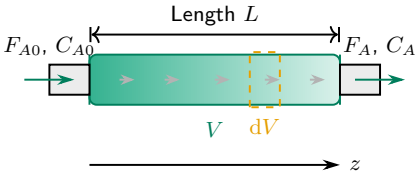


Reactor Schematics: CSTR vs PFR

$$\tau = \frac{C_{A0}\,X}{r(C_{A,\text{exit}})}$$



$$\frac{\mathrm{d}X}{\mathrm{d}\tau} = \frac{r(C_A)}{C_{A0}}$$



CSTR — Well-mixed: $C_A = C_{A,\text{out}}$

- Perfect mixing $\Rightarrow$ uniform C_A
- Exit composition = tank composition
- Good for: highly exothermic reactions (heat removal)

PFR — No axial mixing: $C_A = C_A(z)$

- C_A decreases along length
- Concentration gradient drives reaction
- Good for: high conversion targets

Real-World Application: Three-Way Automotive Catalyst

Monolith reactor ≈ PFR

- Catalyst: Pt–Pd–Rh on cordierite honeycomb
- Simultaneous reactions:
  - $CO + \frac{1}{2} O_2 \rightarrow CO_2$
  - $C_xH_y + O_2 \rightarrow CO_2 + H_2O$
  - $NO_x + CO \rightarrow N_2 + CO_2$
- Target: >95% conversion for **all three**

Why PFR, not CSTR?

At $X > 0.95$, the PFR advantage is **enormous**:
$\tau_{\text{CSTR}}/\tau_{\text{PFR}} = 19/3 \approx 6.3\times$
(first-order, 95% conversion)

Monolith structure

- 400–900 cells per square inch
- Washcoat: high-surface-area γ-Al_2O_3
- Precious metals dispersed on washcoat
- Contact time ∼10 ms at highway speed

A single car catalyst processes ∼100 000 L of exhaust per hour!

Concept Check — Reactor Design

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$(t_{1/2} \propto 1/C_{A0})$

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PFR $(\tau_{\text{PFR}} = 3.0/k$ vs $\tau_{\text{CSTR}} = 19/k)$

From Kinetics to Characterization

Where we stand

- Kinetics tells us **how fast** — rate constants, reaction orders
- TOF normalises to **per-site** activity
- Sabatier principle links **binding strength** to catalytic performance

The missing piece

To calculate TOF, we need the **number of active sites**.
To understand **why** catalysts differ, we need structural information.

Characterization provides the answers.

Next: six complementary techniques, one catalyst.

Outline

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Six Questions, Six Techniques

Question	Technique	Key Output
How much surface area?	BET (N_2 physisorption)	S_{BET} ($\text{m}^2\,\text{g}^{-1}$)
How strongly do species bind?	TPD	E_d ($\text{kJ}\,\text{mol}^{-1}$)
What crystalline phases?	XRD	Phase ID, crystallite size D
What oxidation states?	XPS	Surface composition, valence
How big are the particles?	TEM	$\bar{d}_s$, size distribution
What types of active sites?	CO-FTIR	Site geometry (linear, bridging)

Guiding principle

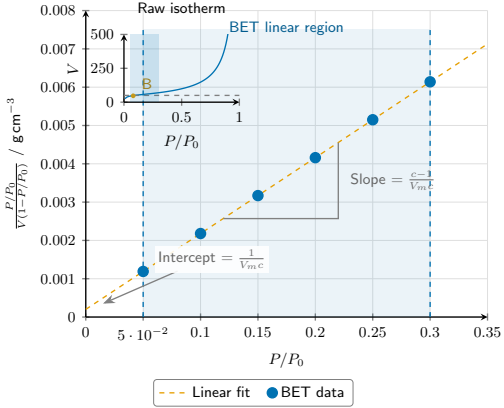
No single technique gives a complete picture. **Complementary** methods are essential.

BET Method: Surface Area from N_2 Adsorption

Linearised BET equation:

$$\frac{P/P_0}{V(1-P/P_0)} = \frac{1}{V_m c} + \frac{c-1}{V_m c} \cdot \frac{P}{P_0}$$

- Plot LHS vs P/P_0 (range 0.05–0.30)
- Slope $= (c-1)/(V_m c)$
- Intercept $= 1/(V_m c)$
- Solve for V_m and c



BET Parameters:
$V_m = 50$ cm^3/g (STP), $c = 100$
Slope $= 0.0198$ g/cm^3, Intercept $= 0.0002$ g/cm^3

Surface area: $S_{\text{BET}} = V_m \times N_A \times \sigma / V_{\text{mol}}$ $(\sigma_{N_2} = 0.162\ \text{nm}^2)$

BET Constant c and Isotherm Shape

$$c = \exp\left(\frac{E_1 - E_L}{RT}\right)$$

Large c (> 100)

- Strong monolayer–surface interaction
- Sharp "knee" in isotherm
- Well-defined monolayer capacity

Small c (< 20)

- Weak interaction with surface
- Gradual isotherm, no clear knee
- Less reliable S_{BET}

Quick rule

$S_{\text{BET}} \approx 4.35 \times V_m$ ($\text{m}^2\,\text{g}^{-1}$) when V_m is in $\text{cm}^3_{\text{STP}}\,\text{g}^{-1}$.

Worked Example: BET Surface Area

Problem

N_2 adsorption on γ-Al_2O_3 at 77 K:

P/P_0	0.05	0.10	0.15	0.20	0.25
V ($cm^3_{STP}\ g^{-1}$)	33.7	37.8	41.5	45.8	50.4
$\frac{P/P_0}{V(1-P/P_0)}$ ($\times 10^{-3}$)	1.56	2.94	4.25	5.46	6.62

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BET plot: slope $= 0.0253$, intercept $= 0.00038$

$$V_m = \frac{1}{0.0253 + 0.00038} = \frac{1}{0.02568} = 38.9\ \text{cm}^3\ \text{g}^{-1}$$

$$c = \frac{0.0253}{0.00038} + 1 \approx 68$$

Surface area:

$$S_{\text{BET}} \approx 4.35 \times 38.9$$

$$= \mathbf{169\ m^2\ g^{-1}}$$

($c = 68 > 20$: reliable result, well-defined monolayer.)

Pore Classification & BJH Method

IUPAC pore classification:

Type	Diameter
Micropores	< 2 nm
Mesopores	2–50 nm
Macropores	> 50 nm

- Zeolites: microporous (∼0.3–1.2 nm)
- γ-Al_2O_3: mesoporous (∼5–15 nm)
- Pellets often have macro + meso

BJH analysis (Barrett–Joyner–Halenda):

Kelvin equation

$$\ln\left(\frac{P}{P_0}\right) = -\frac{2\gamma V_L}{r_k\, RT}$$

(assumes complete wetting, $\theta_c = 0$)

Pores of radius r_k fill at this P/P_0.

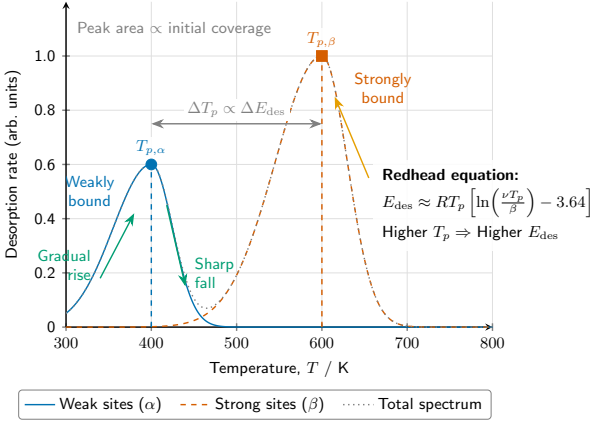
- Use **desorption** branch of isotherm
- Generates pore size distribution: $\mathrm{d}V/\mathrm{d}r$ vs r
- Valid for mesopore range (2–50 nm)

Temperature-Programmed Desorption (TPD)

Redhead equation (1st-order desorption):

$$E_d = RT_p\left[\ln\left(\frac{\nu T_p}{\beta}\right) - 3.64\right]$$

- T_p: peak temperature
- $\nu \approx 10^{13}$ s^{-1} (attempt frequency)
- β: heating rate (K s^{-1}; convert from K/min)



Rule of thumb

E_d (kJ/mol) $\approx 0.28 \times T_p$ (K)

TPR & TPO: Reducibility and Oxidisability

Temperature-Programmed Reduction (TPR):

- Gas: dilute H_2 (5–10% in Ar/N_2)
- Monitor H_2 consumption vs T
- Each peak = a distinct reduction step

Temperature-Programmed Oxidation (TPO):

- Gas: dilute O_2 (2–5% in He)
- Monitor O_2 consumption vs T
- Key application: **coke quantification**

Example: $CuO/ZnO/Al_2O_3$

- Peak at $\sim$220 °C: CuO $\rightarrow$ Cu^0
- Peak area $\propto$ mol Cu reducible
- ZnO: much higher T (not reduced in situ)

Interpretation guide

- Low-T peak (< 350 °C): soft coke (aliphatic)
- High-T peak (> 450 °C): hard coke (graphitic)
- Total area $\rightarrow$ mass of coke deposited

Quick Check — Are You Following?

Concept Check

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2. Two TPD peaks appear at 350 K and 550 K. Which corresponds to the more strongly bound species?

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3. A TPR profile of NiO/Al_2O_3 shows a peak at 600 °C. What does this tell you?

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3. A TPR profile of NiO/Al_2O_3 shows a peak at 600 °C. What does this tell you? **NiO reduces to Ni^0 at 600 °C** (H_2 consumed)

X-ray Diffraction: Phases & Crystallite Size

Bragg's Law (phase identification):

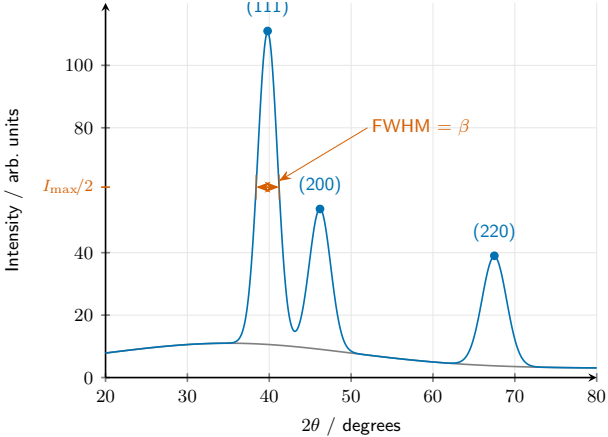
$$n\lambda = 2d\sin\theta$$

Scherrer Equation (crystallite size):

$$D = \frac{K\lambda}{\beta\cos\theta}$$

- $K \approx 0.89$–0.94 (shape factor)
- $\beta =$ FWHM (full width at half maximum) in radians

  (not heating rate—different β!)
- $\lambda_{\text{Cu}\,K_\alpha} = 0.15406$ nm



Bragg: $n\lambda = 2d\sin\theta$ **Scherrer:** $D = K\lambda/(\beta\cos\theta)$, $K \approx 0.9$, $\beta =$ FWHM in rad

Broader peaks $\Rightarrow$ smaller crystallites. Typical catalysts: $D = 2$–50 nm.

Worked Example: Scherrer Crystallite Size

Problem

A Pt/Al_2O_3 catalyst shows a Pt(111) peak at $2\theta = 39.8°$ with FWHM $= 1.2°$. Calculate the crystallite size. ($K = 0.89$, Cu K_α: $\lambda = 0.15406$ nm)

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Step 1: Convert to radians

$$\theta = 39.8°/2 = 19.9°$$

$$\beta = 1.2° \times \frac{\pi}{180°} = 0.02094 \text{ rad}$$

Step 2: Apply Scherrer

$$D = \frac{0.89 \times 0.15406}{0.02094 \times \cos(19.9°)}$$

Step 3: Calculate

$$D = \frac{0.1371}{0.02094 \times 0.9403} = \frac{0.1371}{0.01969} = \mathbf{7.0\ nm}$$

Cross-check with TEM

TEM gives $\bar{d}_s = 6.8$ nm. Good agreement! Scherrer gives **volume-weighted** average.

XPS: Surface Composition & Oxidation States

X-ray Photoelectron Spectroscopy:

$$E_B = h\nu - E_K - \phi$$

- Surface-sensitive (∼1–10 nm depth)
- Identifies **oxidation states** via chemical shifts
- Quantitative surface composition

Chemical shift principle

Higher oxidation state ⇒ fewer valence electrons ⇒ **higher** E_B (core electrons bound more tightly)

Example binding energies

Element	**State**	E_B (eV)
Pt $4f_{7/2}$	Pt^0 / PtO_2	71.2 / 74.5
Pd $3d_{5/2}$	Pd^0 / PdO	335.4 / 336.8
Fe $2p_{3/2}$	Fe^0 / Fe_2O_3	706.7 / 710.9

Practical note

Ultra-high vacuum (UHV) required ($< 10^{-9}$ mbar). Charge correction: C 1s at 284.8 eV.

TEM: Particle Size Distribution & Statistics

TEM workflow:

1. Image nanoparticles at high magnification
2. Measure diameters of **200–300 particles**
3. Compute statistical averages

Three types of average:

Average	Formula
Number: $\bar{d}_n$	$\sum n_i d_i / \sum n_i$
Surface: $\bar{d}_s$	$\sum n_i d_i^3 / \sum n_i d_i^2$
Volume: $\bar{d}_v$	$\sum n_i d_i^4 / \sum n_i d_i^3$

Polydispersity index

$$\text{PDI} = \bar{d}_v / \bar{d}_n$$

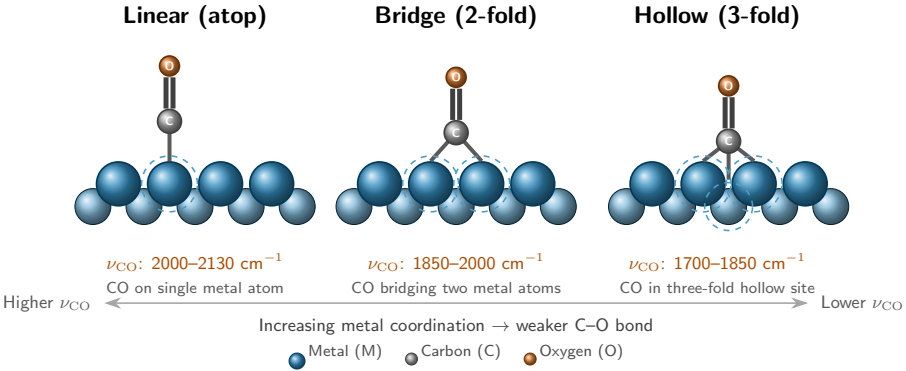
- PDI $= 1.0$: perfectly monodisperse
- PDI < 1.2: narrow distribution
- PDI > 1.5: broad distribution

For catalysis

Use $\bar{d}_s$ (surface-average):

weights by surface area, most relevant to activity.

CO-FTIR: Probing Active Site Geometry



Mode	ν_{CO} (cm^{-1})
Linear (on-top)	2000–2130
Bridging (2-fold)	1850–2000
Hollow (3-fold)	1700–1850

- More bonds to surface ⇒ weaker C–O ⇒ lower ν
- Ratio of modes reveals surface structure

Safety: CO is odourless and highly toxic (TLV-TWA 25 ppm). Always use CO monitors and fume hoods.

Spectroscopy Summary & Complementary Techniques

Reflection-mode FTIR on metal surfaces:

- **Metal surface selection rule:** only vibrations with a dipole moment component **perpendicular** to the surface are IR-active
- On-top CO: strong signal (dipole $\perp$ surface)
- Parallel vibrations: silent on flat metals

Complementary vibrational techniques

- **Raman**: complements IR (different selection rules); good for metal oxides, carbon materials
- **UV-Vis DRS**: electronic transitions, band gap of semiconductors, d–d transitions

Building the full picture

Structural	XRD, TEM
Electronic	XPS, UV-Vis
Vibrational	FTIR, Raman
Thermal	TPD, TPR, TPO
Textural	BET, BJH

A complete characterization study combines at least 3–4 complementary techniques.

Choosing the Right Technique

Information Needed	Primary	Complementary	Sample	Destructive?
Surface area & pores	BET	BJH, t-plot	Powder	No
Binding energies	TPD	TPR/TPO	Powder	No
Crystal phases	XRD	SAED†	Powder	No
Surface oxidation state	XPS	XANES†	Flat/powder	No
Particle size & shape	TEM	SEM, XRD	Powder on grid	No
Site types & geometry	CO-FTIR	CO-TPD	Powder	No

†SAED = Selected Area Electron Diffraction; XANES = X-ray Absorption Near-Edge Structure.

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(broad peak $\Rightarrow$ small D by Scherrer)

3. CO-FTIR shows a strong band at 1920 cm^{-1}. What is the adsorption geometry?

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3. CO-FTIR shows a strong band at 1920 cm^{-1}. What is the adsorption geometry? **Bridging**
(1850–2000 cm^{-1} range)

Integration: Complete Catalyst Characterization

Worked workflow: 2% Pt/γ-Al_2O_3

Step	Technique	Result	Insight
1	BET	$S_{\text{BET}} = 200$ m^2 g^{-1}	High-area mesoporous support
2	XRD (Scherrer)	$D_{\text{Pt}} = 5$ nm	Small Pt nanoparticles
3	TEM	$\bar{d}_s = 4.8$ nm, PDI $= 1.2$	Narrow distribution, consistent with XRD
4	CO-FTIR	70% linear, 30% bridging	Mostly terrace sites
5	CO-TPD	$E_d = 130$ kJ mol^{-1}	Strong CO binding (typical for Pt)

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Dispersion from particle size:

$D \approx 1.13/\bar{d}_s\ (\text{nm}) = 1.13/4.8 = 0.24$

(24% of Pt atoms on surface)

Connect to kinetics:

With $D = 0.24$ and known Pt loading, we can calculate the **TOF** for any measured rate!

Characterization Summary

Quantitative outputs

- S_{BET} from N_2 isotherm
- E_d from TPD (Redhead)
- D from XRD (Scherrer)
- $\bar{d}_s$ from TEM images

Qualitative insights

- XPS: oxidation states
- CO-FTIR: site geometry
- TPR/TPO: reducibility
- SEM: morphology

Complete characterization workflow

BET (area) $\rightarrow$ XRD (phases) $\rightarrow$ TEM (particles) $\rightarrow$ TPD (energetics) $\rightarrow$ CO-FTIR (sites) $\rightarrow$ TOF!

Outline

1 Part I: Reactor Design

2 Part II: Catalyst Characterization

3 Summary & Self-Study

Key Equations Card

Reactor Design:

1. **Batch:** $-\mathrm{d}C_A/\mathrm{d}t = k\,C_A^n$
2. **1st-order:** $\ln(C_A/C_{A0}) = -kt$
3. **TOF:** moles product / (moles sites $\times$ time)
4. **CSTR:** $X = k\tau/(1+k\tau)$
5. **PFR:** $X = 1 - e^{-k\tau}$

Characterization:

6. **BET:** $\frac{P/P_0}{V(1-P/P_0)} = \frac{1}{V_m c} + \frac{c-1}{V_m c}\frac{P}{P_0}$
7. **Redhead:** $E_d \approx 0.28 \times T_p$ (kJ/mol)
8. **Bragg:** $n\lambda = 2d\sin\theta$
9. **Scherrer:** $D = K\lambda/(\beta\cos\theta)$
10. **TEM:** $\bar{d}_s = \sum n_i d_i^3 / \sum n_i d_i^2$

Ten equations—from reactor sizing to crystallite sizing.

NB3: Self-Study Exercises

Exercises (complete before Week 3)

Ex	Topic	Priority
1	Reaction order	Essential
2	TOF calculation	Essential
3	CSTR vs PFR	Essential
4	BET analysis	Essential
5	XRD Scherrer	Bonus

58 cells | ∼2–3 h |
`https://github.com/ExaPsi/2302337`

Self-study from Chapter 3

Topics not covered in today's lecture:

- Differential kinetic methods (§3.3)
- Arrhenius in reactor context (§3.4)
- BJH pore analysis details (§3.6)
- TPR/TPO quantitative analysis (§3.7)
- XPS interpretation (§3.9)

Notebook roadmap

Week 1	NB1, NB2
Week 2	**NB3**
Week 3–5	NB4–NB6

Looking Ahead — Week 3

Next: Transport Limitations (Chapter 4)

- Thiele modulus ϕ and effectiveness factor η
- Internal vs external mass transfer
- Weisz–Prater criterion for diffusion limitation diagnosis

Preparation

- Read Chapter 4, Sections 4.1–4.3
- Complete NB3 essential exercises
- Review BET surface area concept

Key question for Week 3

If a catalyst pellet is "too large", how much activity do we lose to **diffusion limitations**?

See you next week!